

Loss of complex female song but not duetting in the ancestors of Carolina wrens

Supplemental Material

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TABLE S1. Character scores and source information for 42 wren species included in phylogenetic comparative analyses.

Species	Female vocal complexity	Vocal coordination	Breeding latitude	References
<i>Campylorhynchus chiapensis</i>	Complex	Coordinated	Tropical	Winkler et al. 2020
<i>Campylorhynchus fasciatus</i>	Complex	Coordinated	Tropical	Winkler et al. 2020
<i>Campylorhynchus griseus</i>	Complex	Coordinated	Tropical	Winkler et al. 2020
<i>Campylorhynchus jocosus</i>	Complex	Coordinated	Tropical	Winkler et al. 2020
<i>Campylorhynchus megalopterus</i>	Complex	Coordinated	Tropical	Winkler et al. 2020
<i>Campylorhynchus nuchalis</i>	Complex	Coordinated	Tropical	Winkler et al. 2020
<i>Campylorhynchus rufinucha</i>	Complex	Coordinated	Tropical	Bradley and Mennill 2009; Winkler et al. 2020
<i>Campylorhynchus yucatanicus</i>	Complex	Coordinated	Tropical	Winkler et al. 2020
<i>Campylorhynchus zonatus</i>	Complex	Coordinated	Tropical	Winkler et al. 2020
<i>Cantorchilus guarayanus</i>	Complex	Coordinated	Tropical	Mann et al. 2009; Winkler et al. 2020
<i>Cantorchilus leucotis</i>	Complex	Coordinated	Tropical	Mann et al. 2009; Winkler et al. 2020
<i>Cantorchilus longirostris</i>	Complex	Coordinated	Tropical	Winkler et al. 2020
<i>Cantorchilus modestus</i>	Complex	Coordinated	Tropical	Cuthbert and Mennill 2007; Mann et al. 2009; Winkler et al. 2020
<i>Cantorchilus nigricapillus</i>	Complex	Coordinated	Tropical	Mann et al. 2009; Levin 1996; Winkler et al. 2020
<i>Cantorchilus semibadius</i>	Complex	Coordinated	Tropical	Mann et al. 2009; Winkler et al. 2020
<i>Cantorchilus superciliaris</i>	Complex	Coordinated	Tropical	Mann et al. 2009; Winkler et al. 2020
<i>Cantorchilus thoracicus</i>	Complex	Coordinated	Tropical	Mann et al. 2009; Winkler et al. 2020
<i>Catherpes mexicanus</i>	Complex	Uncoordinated	Temperate	Hathcock & Benedict 2018; Winkler et al. 2020
<i>Cinnycerthia peruana</i>	Complex	Coordinated	Tropical	Winkler et al. 2020
<i>Cistothorus palustris</i>	Simple	Uncoordinated	Temperate	Winkler et al. 2020
<i>Cistothorus platensis</i>	Simple	Uncoordinated	Temperate	Winkler et al. 2020
<i>Cyphorhinus arada</i>	Complex	Coordinated	Tropical	Winkler et al. 2020
<i>Henicorhina leucophrys</i>	Complex	Coordinated	Tropical	Winkler et al. 2020
<i>Hylorchilus sumichrasti</i>	Simple	Uncoordinated	Tropical	Pérez-Villafañe et al. 1999; deSilva et al. 2004; Winkler et al. 2020
<i>Pheugopedius atrogularis</i>	Complex	Coordinated	Tropical	Mann et al. 2009; Winkler et al. 2020

<i>Pheugopedius coraya</i>	Complex	Coordinated	Tropical	Mann et al. 2009; Winkler et al. 2020
<i>Pheugopedius euophrys</i>	Complex	Coordinated	Tropical	Mann et al. 2009; Winkler et al. 2020; Coleman et al. 2021
<i>Pheugopedius fasciatoventris</i>	Complex	Coordinated	Tropical	Mann et al. 2009; Winkler et al. 2020
<i>Pheugopedius felix</i>	Complex	Coordinated	Tropical	Mann et al. 2009; Templeton et al. 2013; Winkler et al. 2020
<i>Pheugopedius genibarbis</i>	Complex	Coordinated	Tropical	Mann et al. 2009; Winkler et al. 2020
<i>Pheugopedius maculipectus</i>	Complex	Coordinated	Tropical	Mann et al. 2009; Winkler et al. 2020
<i>Pheugopedius mystacalis</i>	Complex	Coordinated	Tropical	Mann et al. 2009; Winkler et al. 2020
<i>Pheugopedius rutilus</i>	Complex	Coordinated	Tropical	Mann et al. 2009; Winkler et al. 2020
<i>Pheugopedius sclateri</i>	Complex	Coordinated	Tropical	Mann et al. 2009; Winkler et al. 2020
<i>Salpinctes obsoletus</i>	Simple	Uncoordinated	Temperate	Winkler et al. 2020
<i>Thryophilus pleurostictus</i>	Complex	Uncoordinated	Tropical	Mann et al. 2009; Molles and Vehrencamp 1999; Winkler et al. 2020
<i>Thryophilus rufalbus</i>	Complex	Coordinated	Tropical	Mennill and Vehrencamp 2005; Mann et al. 2009; Masco et al. 2016; Winkler et al. 2020
<i>Thryophilus sinaloa</i>	Complex	Uncoordinated	Tropical	Mann et al. 2009; Winkler et al. 2020
<i>Thryothorus ludovicianus</i>	Simple	Coordinated	Temperate	Shuler 1965; Benedict 2008; Mann et al. 2009; Winkler et al. 2020
<i>Troglodytes aedon</i>	Complex	Both	Both	Johnson and Kermott 1990; Mann et al. 2009; Krieg and Getty 2016; Winkler et al. 2020
<i>Troglodytes sissonii</i>	Complex	Coordinated	Tropical	Howell and Webb 1995; Baptista and Martínez Gómez 2002; Winkler et al. 2020
<i>Uropsila leucogastra</i>	Complex	Coordinated	Tropical	Winkler et al. 2020

TABLE S1: LITERATURE CITED

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TABLE S2. Effect sizes calculated as Cohen's W for tests of vocal coordination by female Carolina wrens with respect to males.

Pair	SampleGaps	KeepGaps	KeepSongOrder	Duty Cycle
1	0.02	0.02	0.00	0.05
2	0.32	0.33	0.29	0.20
3	0.04	0.04	0.09	0.06
3	0.36	0.35	0.39	0.28
4	0.24	0.26	0.23	0.13
4	0.21	0.20	0.19	0.09
4	0.01	0.01	0.01	0.08
5	0.07	0.15	0.11	0.16
5	0.23	0.15	0.17	0.38
5	0.19	0.20	0.22	0.45
6	0.18	0.16	0.14	0.47
7	0.11	0.17	0.12	0.07
8	0.03	0.03	0.04	0.02
9	0.53	0.44	0.59	0.41

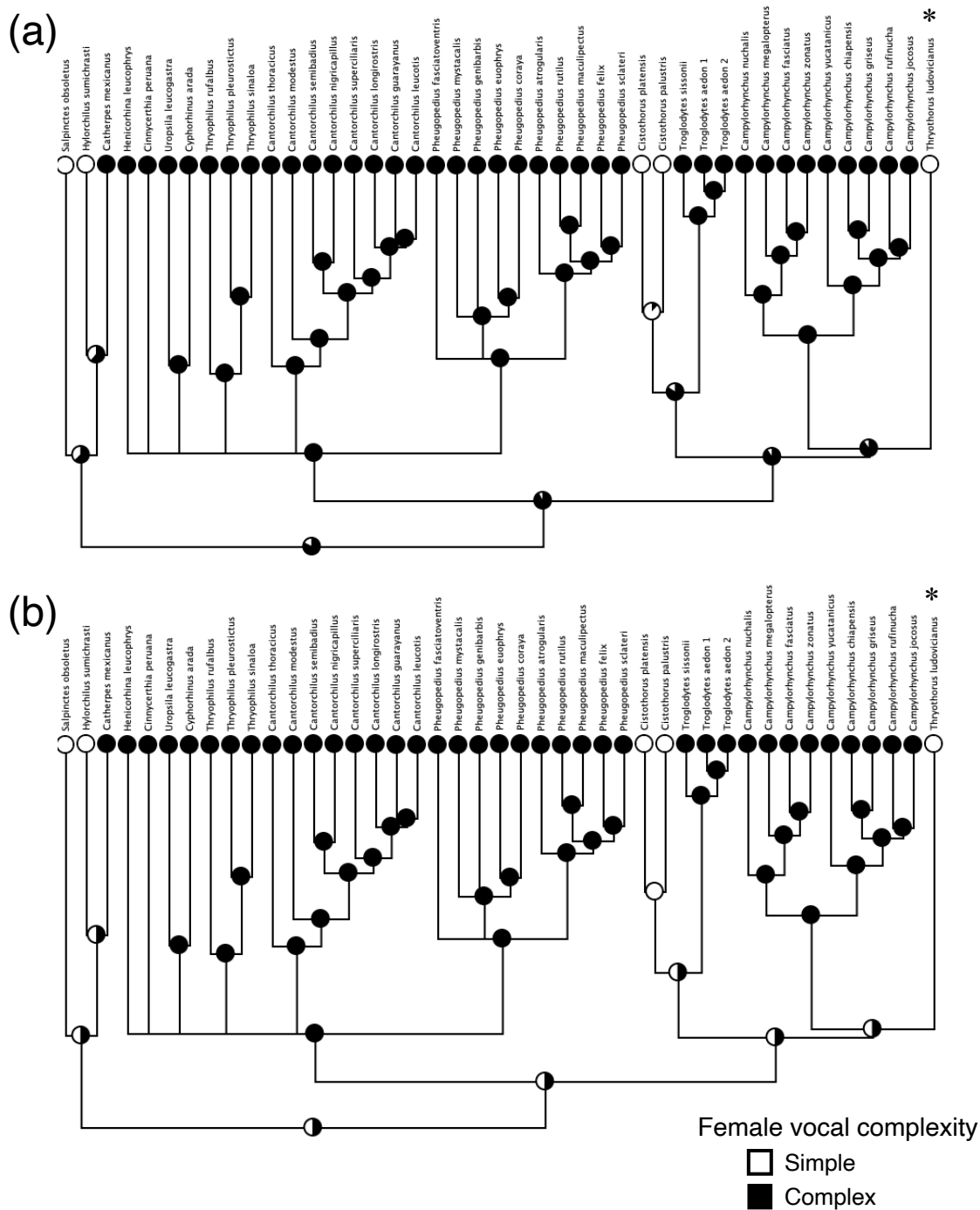


FIGURE S1. Ancestral character state reconstructions of female vocal complexity on a phylogeny of 42 wren species, with black and white fill showing (a) proportional likelihoods from Markov k-state one parameter maximum-likelihood analysis or (b) character states resolved using unordered parsimony. Carolina wren (*Thyrothorus ludovicianus*) is indicated by an asterisk.

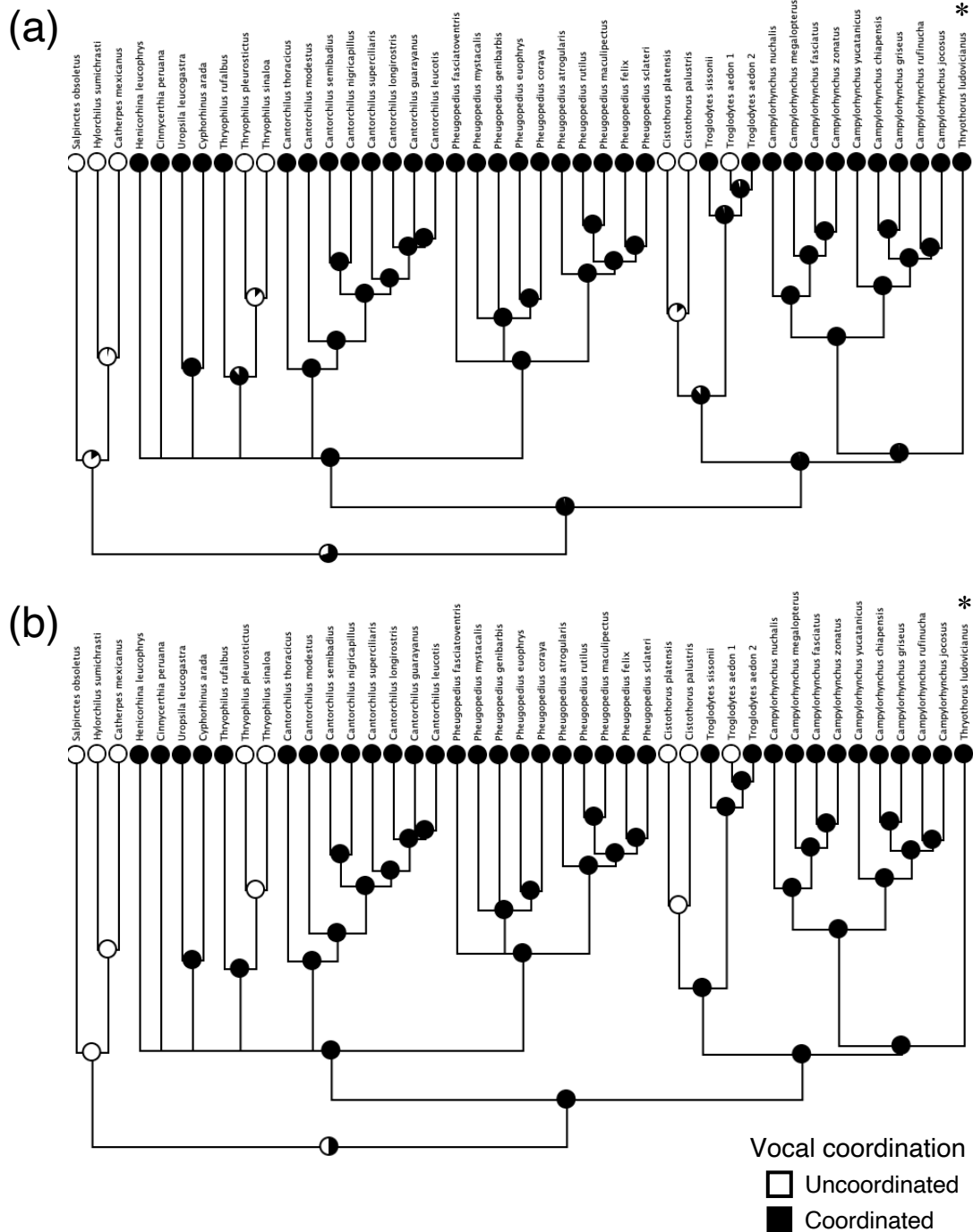


FIGURE S2. Ancestral character state reconstructions of vocal coordination between males and females on a phylogeny of 42 wren species, with black and white fill showing (a) proportional likelihoods from Markov k-state one parameter maximum-likelihood analysis or (b) character states resolved using unordered parsimony. Carolina wren (*Thyothorus ludovicianus*) is indicated by an asterisk.

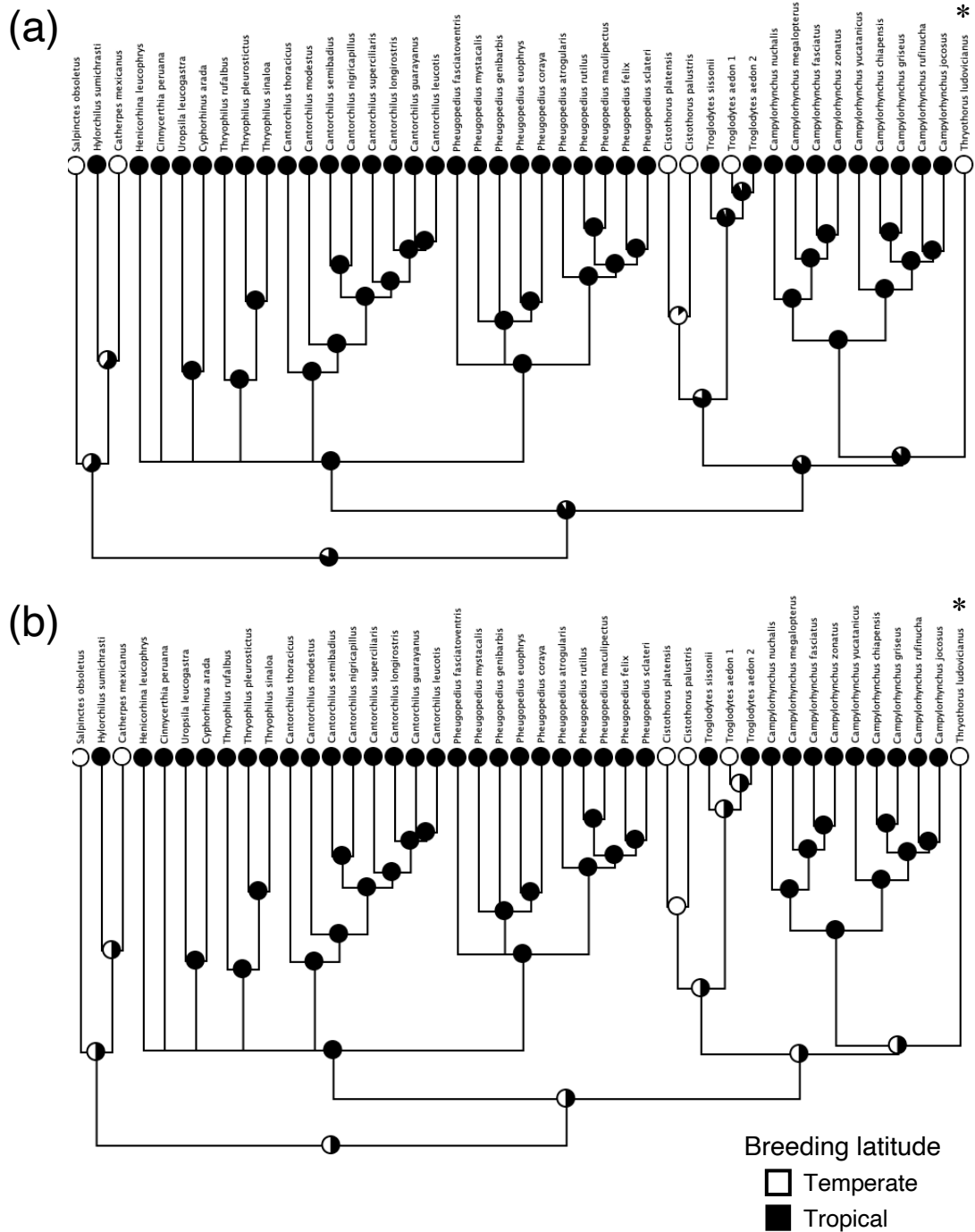


FIGURE S3. Ancestral character state reconstructions of breeding latitude on a phylogeny of 42 wren species, with black and white fill showing (a) proportional likelihoods from Markov k-state one parameter maximum-likelihood analysis or (b) character states resolved using unordered parsimony. Carolina wren (*Thyrothorus ludovicianus*) is indicated by an asterisk.